

Final model · Stage 1: model choice and calibration

Finalizing the credit-risk engine · validation split only · test still sealed

Result: the final model is **M3-B-mono** (LightGBM with monotonic constraints), calibrated with **Platt scaling**. The constraints cost only 0.0008 ROC-AUC, yet they remove 3,709 cases where the unconstrained model broke basic credit logic. Calibration lifts the average PD from 18.70% to 20.35%, matching the real 20.35% default rate.

1. Integrity check

Both models were trained on Kaggle (LightGBM 4.6) and reloaded locally (LightGBM 4.7.0). Their validation PDs match the predictions saved on Kaggle to within 1e-16, so the saved model files are exactly the models we evaluated.

2. Do the monotonic constraints really hold?

For 1,000 random validation applicants, each rule's feature was swept across its 1st-99th percentile range (up to 25 values) while everything else stayed fixed. An applicant counts as a violation if their PD ever moves the wrong way.

Rule the model must follow	M3-B: applicants violating	M3-B: worst wrong move	M3-B-mono: applicants violating
Higher FICO → lower PD	676 / 1,000 (68%)	2.08 PD points	0 / 1,000
Higher DTI → higher PD	963 / 1,000 (96%)	0.96 PD points	0 / 1,000
Higher income → lower PD	999 / 1,000 (100%)	4.20 PD points	0 / 1,000
60-month term → higher PD	0 / 1,000 (0%)	0.00 PD points	0 / 1,000
More recent inquiries → higher PD	71 / 1,000 (7%)	0.40 PD points	0 / 1,000
Higher card utilisation → higher PD	1000 / 1,000 (100%)	1.52 PD points	0 / 1,000

- The unconstrained model breaks credit logic for almost everyone somewhere: for nearly every applicant there is an income range where **earning more raises their PD** (by up to 4.2 points). These wiggles are noise the trees picked up, not real risk, and they would be impossible to defend to a customer or regulator.
- The monotonic model has **zero** violations across all six rules, for a cost of only 0.0008 ROC-AUC (0.7362 → 0.7354).

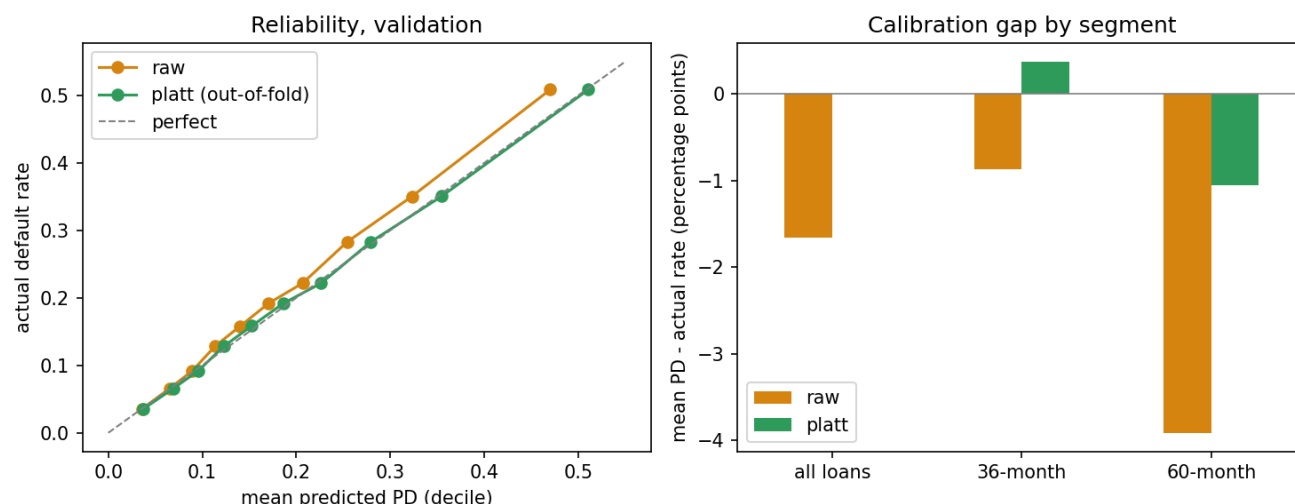
3. Model choice

Rule fixed in advance: *B-mono if ROC-AUC cost < 0.005 and 0 monotonicity violations, else B*. B-mono costs 0.0008 (< 0.005) and has 0 violations, so **M3-B-mono** is the final model (147 features, no LendingClub grade features).

4. Calibration: making the PD an honest probability

Raw LightGBM PDs run too low on 2015 loans (the drift seen since M0). Three options were compared **out-of-fold**: 5-fold cross-fitting inside validation, so no loan's calibrated PD comes from a calibrator that saw that loan.

Method	ROC-AUC	Brier	Log loss	Mean PD (actual 20.35%)	Calibration error (ECE)	Gap 36-month	Gap 60-month
Raw LightGBM PD	0.7354	0.14277	0.4467	18.70%	0.0166	-0.87 pts	-3.91 pts
Platt scaling	0.7354	0.14236	0.4456	20.35%	0.0041	+0.37 pts	-1.06 pts
Isotonic regression	0.7347	0.14245	0.4460	20.35%	0.0015	+0.38 pts	-1.10 pts



Rule: lowest out-of-fold Brier score; Platt wins ties within 0.00005 because it is simpler. **Platt** was chosen (Brier 0.14277 → 0.14236).

The calibrator (saved as plain JSON, no pickle): $PD_{cal} = 1 / (1 + e^{-(a \cdot \logit(PD_{raw}) + b)})$ with **a = 1.0412** and **b = 0.1723**. With $a \approx 1$, it keeps the model's spread and mainly shifts all PDs up to the 2015 default level. Because it is monotone, the ranking (ROC-AUC) does not change.

What we learned

- **Calibration error falls about 4×** (ECE 0.0166 → 0.0041); the mean PD now matches the actual default rate.
- Isotonic is even closer per decile (ECE 0.0015) but scores a slightly worse Brier and log loss and lowers ROC-AUC to 0.7347 (its step function creates ties), so Platt is the better trade-off.
- **Remaining gap:** 60-month loans are still under-predicted by 1.1 points and 36-month loans over-predicted by 0.4. One global calibrator cannot fix both; a term-specific calibrator is an option if the test set confirms the gap.
- **Honesty note:** validation has now been used to stop boosting, choose settings and fit the calibrator. Only the sealed test set (Stage 5) can give an unbiased final score; that is why it stays closed.

Next: Stage 2, turning SHAP values into plain-language reason codes, plus fairness checks across states and zip regions.